# Mathematics of the Vacuum Tape Simulator

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## 1 Introduction

The Vacuum Tape Simulator is a digital signal processing (DSP) plugin designed to emulate the non-linear characteristics of vacuum tube amplification and magnetic tape recording. This document details the mathematical models used for compression, hysteresis, frequency response, and mechanical transport imperfections.

## 2 Vacuum Tube Compression (Voltage Drain Model)

The vacuum tube stage is modeled using a "Voltage Drain" approach. The system maintains an internal state variable $V[n]$, representing the available headroom (voltage).

The input signal $x[n]$ is used to calculate the energy $E[n]$:

$$E[n] = |x[n]| \tag{1}$$

The voltage $V[n]$ decays based on the energy exceeding a threshold $T$:

$$V[n] = V[n-1] + \alpha \cdot \text{drain} \cdot \max(0, E[n] - T) + \beta \cdot \text{recovery} \cdot (1 - V[n-1]) \tag{2}$$

where $\alpha$ and $\beta$ are time constants derived from the sample rate. The output signal $y[n]$ is then:

$$y[n] = x[n] \cdot V[n] \tag{3}$$

## 3 Magnetic Tape Hysteresis

Tape saturation is modeled using a stateful hysteresis function that accounts for the magnetic "memory" of the tape particles. We use a modified tanh model:

$$y[n] = \tanh(x[n] \cdot G - h \cdot \tanh((x[n] - x[n-1]) \cdot \beta)) \tag{4}$$

where:

- $G$ is the Tape Drive (gain).
- $h$ is the Hysteresis coercivity (lag amount).
- $\beta$ is the smoothing factor for the derivative.

## 4 Frequency-Dependent Losses (Tape Speed)

Tape speed affects the high-frequency response (Wallace Losses). This is modeled using a first-order IIR low-pass filter:

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}} \tag{5}$$

The cutoff frequency $f_c$ is mapped from the tape speed $S$ (in ips):

$$f_c = 5000 + \frac{S}{30} \cdot 13000 \tag{6}$$

## 5 Mechanical Transport (Wow and Flutter)

Wow (low-frequency) and Flutter (high-frequency) are modeled as time-varying delay modulations $\Delta t[n]$.

The delay time is modulated by a combination of a slow "drunk walk" LFO ($L_{wow}$) and random noise ($N_{flutter}$):

$$\Delta t[n] = \text{Wow} \cdot L_{wow}[n] + \text{Flutter} \cdot N_{flutter}[n] \tag{7}$$

The signal is then retrieved using cubic interpolation from a fractional delay line.